PRACTICAL-4

Problem Definition

Add dynamic UI components such as collapsible FAQ, modal popup,image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

1.How are DOM elements selected and modified?

DOM elements are selected using methods like getElementById(), querySelector(), and querySelectorAll(). They can be modified using properties like innerHTML, textContent, style, and classList

2.Are event listeners attached correctly?

Yes. Event listeners are attached using addEventListener(), which allows JavaScript to respond to events such as click, submit, change, etc.

3.Is localStorage used for remembering theme choice?

Yes. localStorage can store the user's selected light/dark theme so that the same theme is applied when the page is opened again.

4.Does interactivity improve usability without breaking accessibility?

Yes. Features like menus, modals, notifications, and theme switching improve usability. They should also provide clear controls, readable text, keyboard access, and proper labels so accessibility is maintained.

```
function login() {

    let username = document.getElementById("username").value;

    let password = document.getElementById("password").value;

    let usernamePattern = /^[0-9]{2}[A-Za-z]{3}[0-9]{3,4}$/;

    let passwordPattern =
/^(?=.*[A-Za-z])(?=.*[0-9])(?=.*@)[A-Za-z0-9@]{8,}$/;

    if (username == "" || password == "") {

        alert("Please enter username and password");

    }

    else if (!usernamePattern.test(username)) {

        alert("Invalid Username");

    }
```

```
    else if (!passwordPattern.test(password)) {

        alert("Invalid Password");

    }

    else {

        alert("Login Successful!");

  window.location.href = "dashboard.html";

    }

}

// PAGE LOAD

window.onload = function() {

// FAQ

let faq = document.createElement("button");

faq.innerHTML = "What is Student Hub?";

document.body.appendChild(faq);

let answer = document.createElement("p");

answer.innerHTML = "Student Hub helps students manage college
information.";

answer.style.display = "none";

document.body.appendChild(answer);

faq.onclick = function() {

    if (answer.style.display == "none") {

        answer.style.display = "block";

    }

    else {

        answer.style.display = "none";
```

```
    }
};
  // 3. NOTIFICATION BANNER
setTimeout(function() {
    let notification = document.createElement("div");
    notification.innerHTML = "🔔 New Notice: Check the Notice Board!";
    notification.style.backgroundColor = "#4a90e2";
    notification.style.color = "white";
    notification.style.padding = "10px";
    notification.style.margin = "15px";
    document.body.appendChild(notification);
}, 2000);
    // 4. HAMBURGER MENU
    let menuButton = document.createElement("button");
    menuButton.innerHTML = "☰ Menu";
    menuButton.style.position = "fixed";
    menuButton.style.top = "10px";
    menuButton.style.right = "190px";
    menuButton.style.backgroundColor = "#4a90e2";
    menuButton.style.color = "white";
    document.body.appendChild(menuButton);
    let navigation = document.querySelector("table");
    navigation.style.display = "none";
    menuButton.onclick = function() {
        if (navigation.style.display == "none") {
```

```
            navigation.style.display = "table";

        }

        else {

            navigation.style.display = "none";

        }

    };

    // 5. DARK MODE

    let darkButton = document.createElement("button");

    darkButton.innerHTML = "Dark Mode";

    darkButton.style.backgroundColor = "#4a90e2";

    darkButton.style.color = "white";

    darkButton.style.position = "fixed";

    darkButton.style.top = "10px";

    darkButton.style.right = "100px";

    document.body.appendChild(darkButton);

    darkButton.onclick = function() {

        document.body.style.backgroundColor = "black";

        document.body.style.color = "white";

    };

    // 6. LIGHT MODE

    let lightButton = document.createElement("button");

    lightButton.innerHTML = "Light Mode";

    lightButton.style.backgroundColor = "#4a90e2";

    lightButton.style.color = "white";

    lightButton.style.position = "fixed";
```

```
    lightButton.style.top = "10px";

    lightButton.style.right = "10px";

    document.body.appendChild(lightButton);

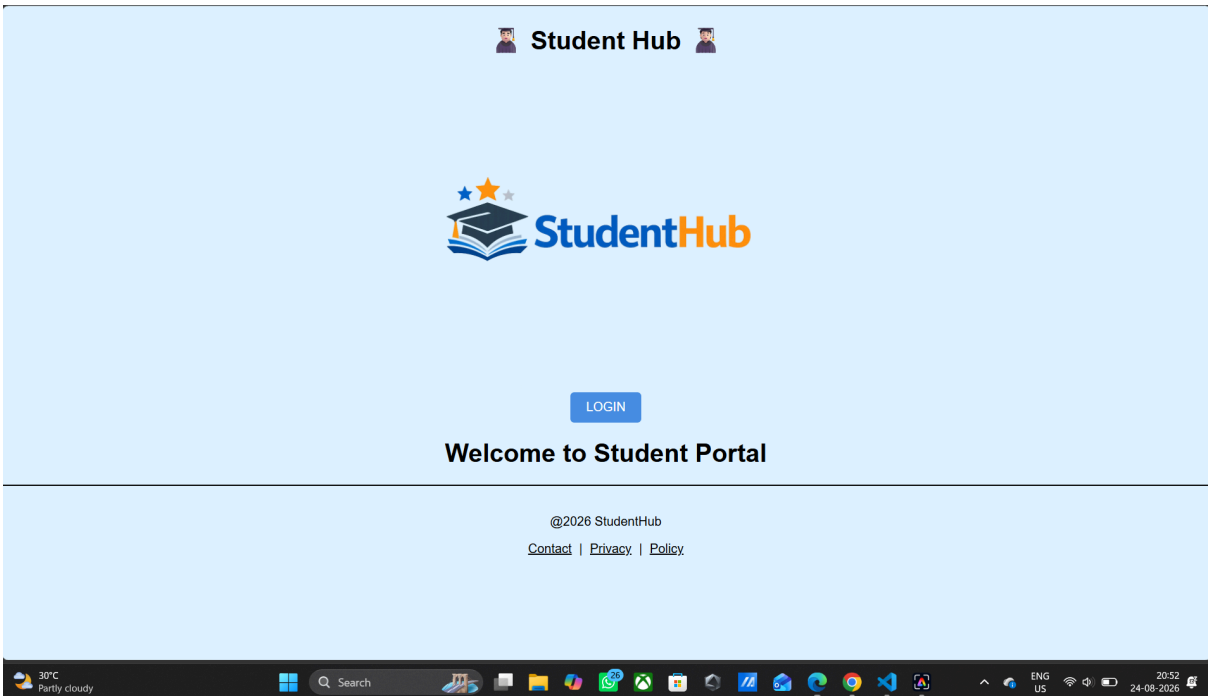
    lightButton.onclick = function() {

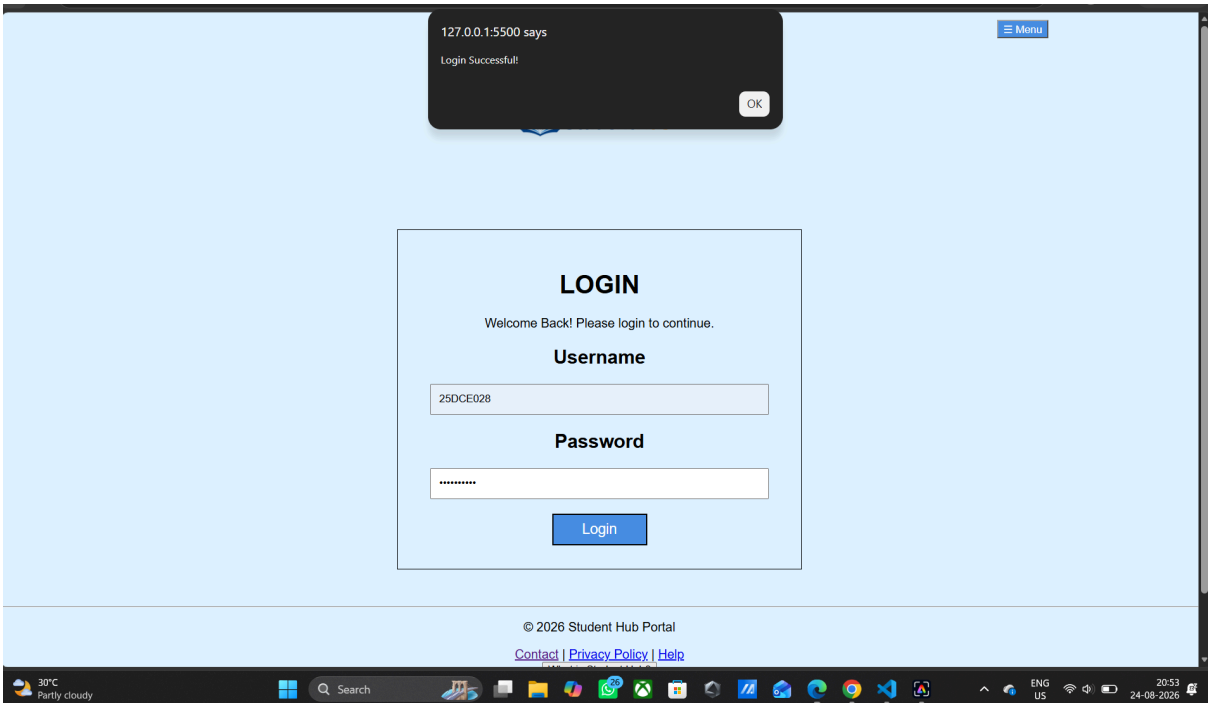
        document.body.style.backgroundColor = "#dff3ff";

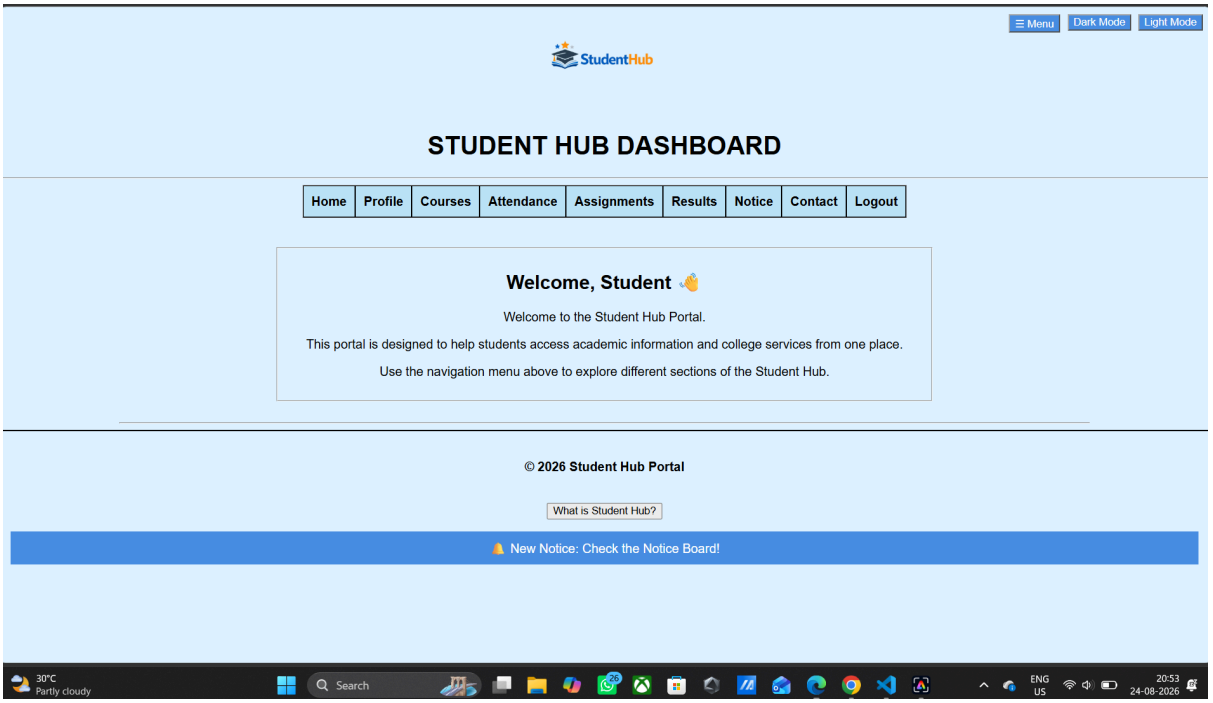
        document.body.style.color = "black";

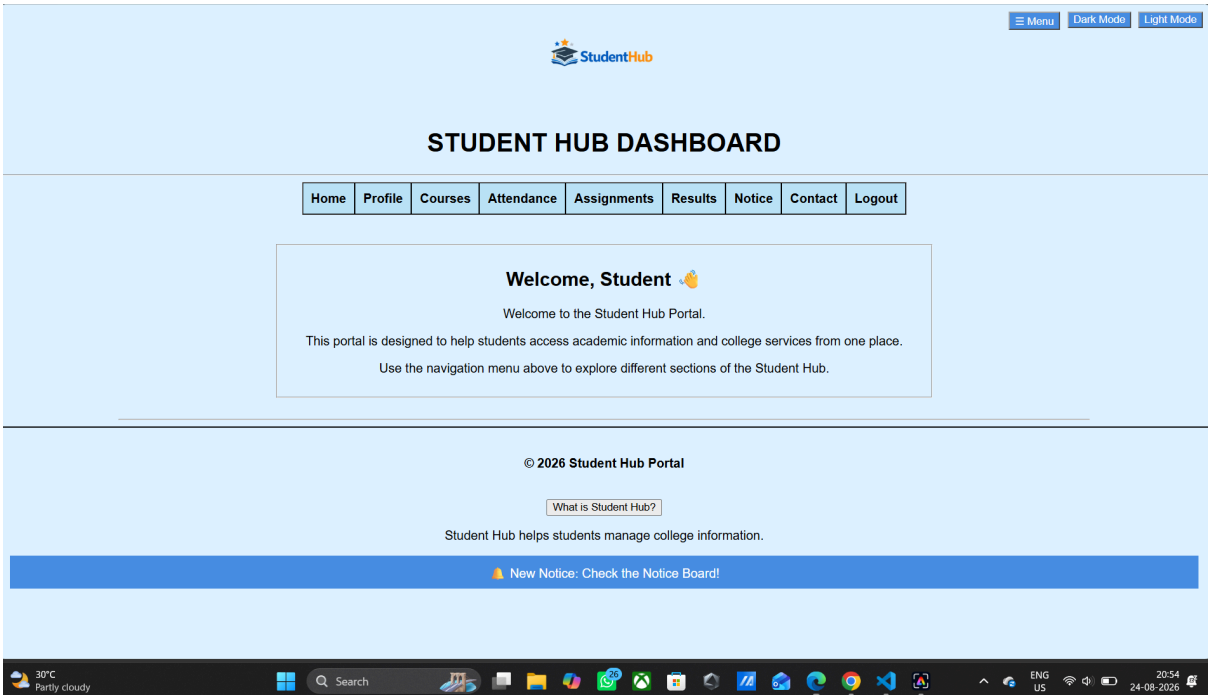
    };

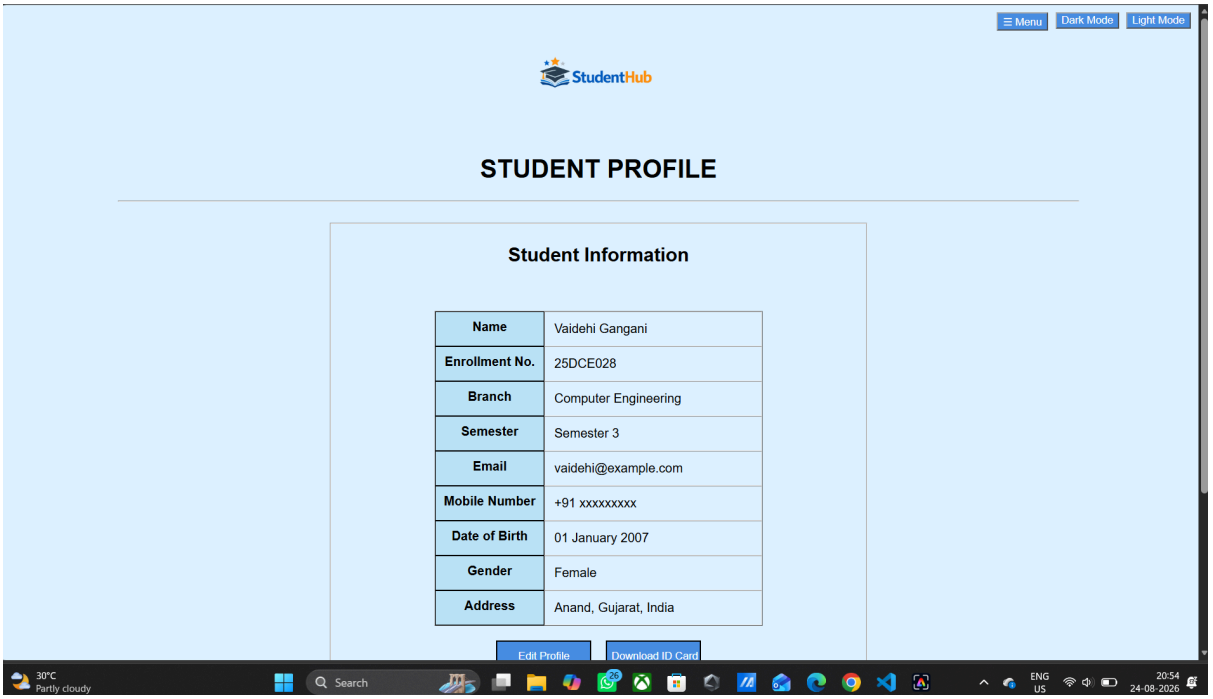
};
```

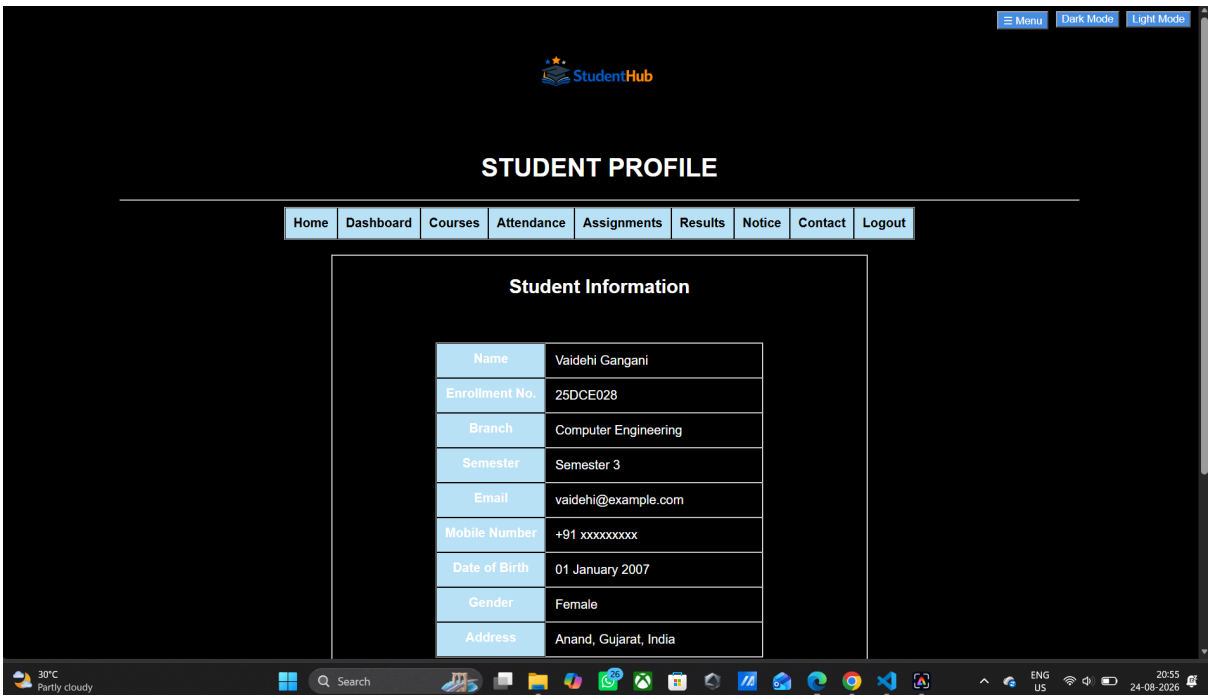


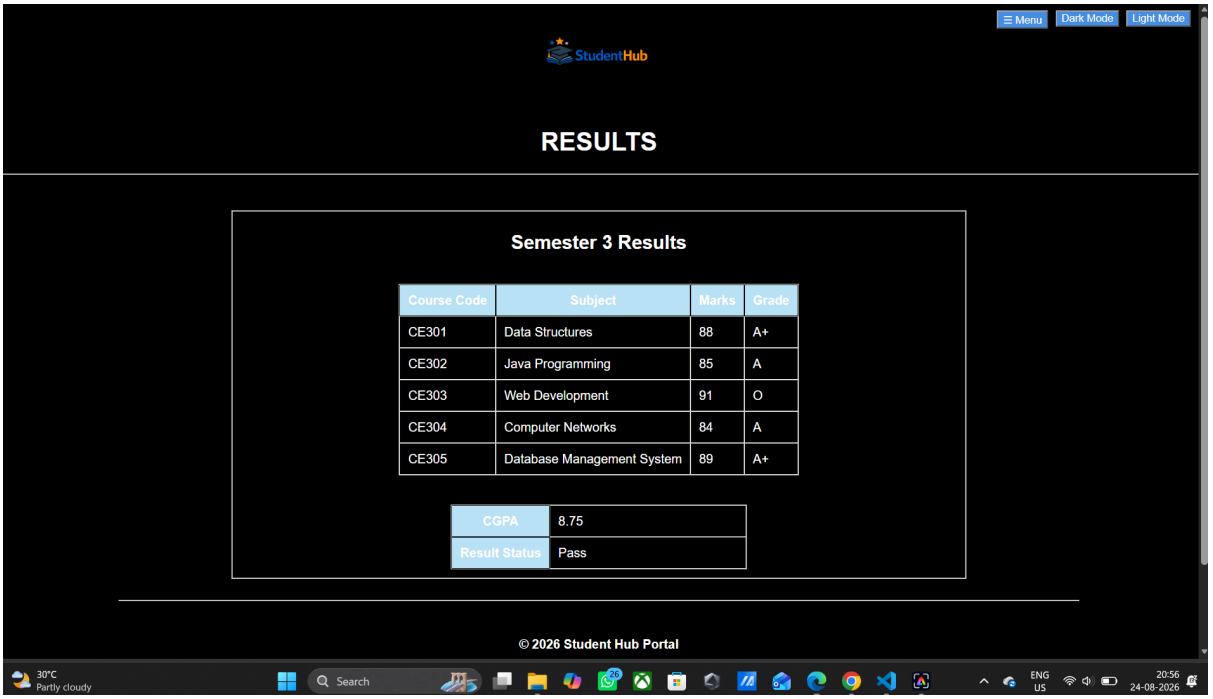

127.0.0.1:5500 says
Login Successful!
OK
☰ Menu
LOGIN
Welcome Back! Please login to continue.
Username
25DCE028
Password
Login
© 2026 Student Hub Portal
Contact | Privacy Policy | Help

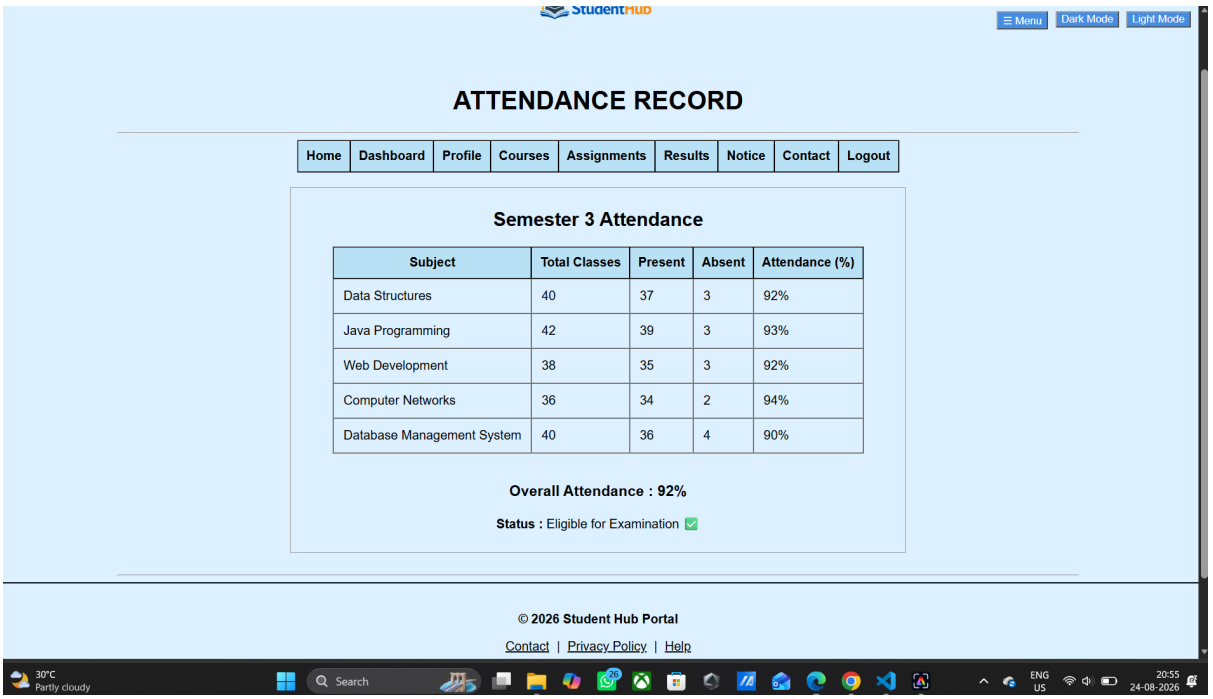

☰ Menu
Dark Mode
Light Mode
StudentHub
STUDENT HUB DASHBOARD
Home
Profile
Courses
Attendance
Assignments
Results
Notice
Contact
Logout
Welcome, Student
Welcome to the Student Hub Portal.
This portal is designed to help students access academic information and college services from one place.
Use the navigation menu above to explore different sections of the Student Hub.
© 2026 Student Hub Portal
What is Student Hub?
New Notice: Check the Notice Board!


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© 2026 Student Hub Portal
What is Student Hub?
Student Hub helps students manage college information.
New Notice: Check the Notice Board!
30°C
Partly cloudy
Search
ENG
US
20:54
24-08-2026


Menu
Dark Mode
Light Mode
StudentHub
STUDENT PROFILE
Student Information
Name
Vaidehi Gangani
Enrollment No.
25DCE028
Branch
Computer Engineering
Semester
Semester 3
Email
vaidehi@example.com
Mobile Number
+91 xxxxxxxxx
Date of Birth
01 January 2007
Gender
Female
Address
Anand, Gujarat, India
Edit Profile
Download ID Card
30°C
Partly cloudy
Search
ENG
US
20:54
24-08-2026


Menu
Dark Mode
Light Mode
StudentHub
STUDENT PROFILE
Home
Dashboard
Courses
Attendance
Assignments
Results
Notice
Contact
Logout
Student Information
Name
Vaidehi Gangani
Enrollment No.
25DCE028
Branch
Computer Engineering
Semester
Semester 3
Email
vaidehi@example.com
Mobile Number
+91 xxxxxxxxx
Date of Birth
01 January 2007
Gender
Female
Address
Anand, Gujarat, India


Menu
Dark Mode
Light Mode
StudentHub
RESULTS
Semester 3 Results
Course Code
Subject
Marks
Grade
CE301
Data Structures
88
A+
CE302
Java Programming
85
A
CE303
Web Development
91
O
CE304
Computer Networks
84
A
CE305
Database Management System
89
A+
CGPA
8.75
Result Status
Pass
© 2026 Student Hub Portal





Implementation

- Created dynamic UI components using JavaScript.
- Implemented collapsible FAQ to show/hide answers.
- Added a modal popup that opens and closes on button clicks.
- Added a notification banner for user messages.
- Implemented a hamburger menu for navigation.
- Added a light/dark theme switcher using JavaScript.
- Used DOM manipulation and event handling to make the UI interactive.

Challenges

- Handling click events for different UI components.
- Selecting and modifying DOM elements correctly.
- Managing show/hide functionality for FAQ, modal, and menu.
- Applying light and dark themes consistently.
- Making the components responsive and user-friendly.

GitHub Repository Link:https://github.com/Vaidehi137/student-hub-portal

Conclusion :

The website successfully uses HTML, CSS, and JavaScript to create an interactive and user-friendly interface. DOM manipulation and event listeners provide dynamic functionality, while localStorage remembers the user's theme preference. Interactive components such as the menu, modal, notification, and slider improve usability while maintaining accessibility. Overall, the project demonstrates how JavaScript can make a web page more interactive, responsive, and user-friendly.